Single-cell profiling of frog tail regeneration identifies ROC markers

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Abstract

I analyzed 13,199 single cells from *Xenopus laevis* tails across developmental stages (st40, st46) and 0–3 days post-amputation. After log-normalization and selection of 2,300 highly variable genes, I built a PCA–kNN–UMAP graph and clustered cells with Leiden and Louvain. The two partitions agreed well (ARI = 0.755; NMI = 0.891); I used the Leiden solution at resolution 0.4 (silhouette on PCA = 0.275) for downstream work. An allele-aware composite score for *tp63*, *lef1*, and keratin genes highlighted a single epidermal cluster (cluster 15) as enriched for the Regenerative Organizing Cell (ROC). Differential expression (Wilcoxon) and logistic-regression analyses showed substantial agreement among top markers (32/50 overlap; Jaccard = 0.471). Two denoising approaches (MAGIC and PCA low-rank) and two batch-integration methods (Harmony and BBKNN) modestly altered cluster compactness but preserved ROC localization and marker stability. Comparison of my ROC gene list to Supplementary Table 3 showed significant enrichment despite L/S allele differences (strict overlap 2/100; $p = 2.7 \times 10^{-3}$; allele-collapsed overlap 5; Jaccard = 0.062). Overall, these results position the ROC within epidermis and provide a reproducible workflow for its identification.

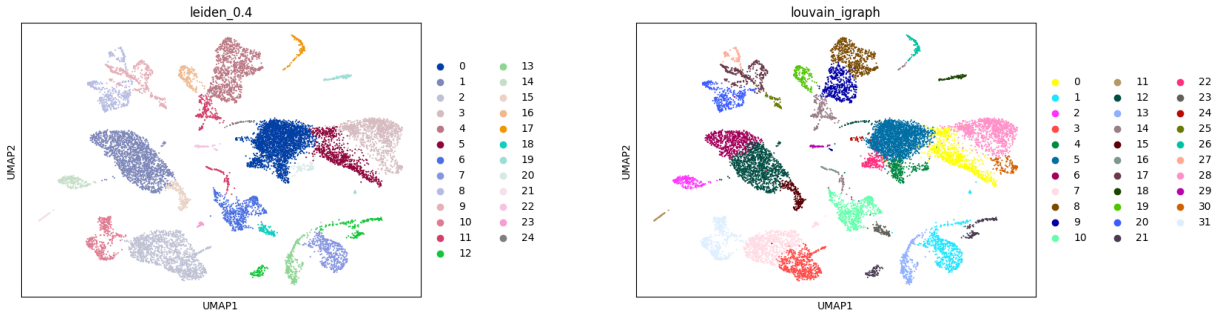

Figure 1. UMAP visualization of Leiden clustering (resolution 0.4) showing major tissue compartments of the regenerating frog tail.

Introduction

The frog *Xenopus laevis* can regenerate its tail during defined developmental windows, providing a tractable vertebrate model for understanding tissue repair. Single-cell RNA sequencing (scRNA-seq) enables this process to be examined at cellular resolution, revealing coordinated activity among epidermal, mesenchymal, immune, and neural lineages. Within the regenerating epidermis, a small population termed the Regenerative Organizing Cell (ROC) has been proposed to orchestrate wound closure and signaling cues. However, reliably pinpointing this population in pooled datasets that span stages and timepoints remains challenging, particularly given batch effects and allele-specific gene annotations (`.L`/`.S`) common in *Xenopus*.

Here, I address the practical problem of locating the ROC in a public scRNA-seq atlas of the frog tail and defining the gene program that distinguishes it from neighboring epidermal states. I also compare the resulting gene set to those reported previously to assess concordance and biological reproducibility.

Methods

Data

I analyzed the processed dataset `cleaned_processed_frogtail.h5ad`, containing 13,199 cells and 31,535 genes from *Xenopus laevis* tails across intact and 0–3 days post-amputation conditions (stages 40 and 46). Metadata included `sample`, `DevelopmentalStage`, `DaysPostAmputation`, `Condition`, and `batch` (1–4). All analyses were implemented in Python using Scanpy 1.10.2 and AnnData 0.12.2.

Preprocessing

Raw counts were retained in `adata.layers["counts"]`. I performed library-size normalization (10,000 counts/cell), log-transformation, and selected 2,300 highly variable genes. Cells with < 200 detected genes or high mitochondrial percentage were removed.

Dimensionality reduction and clustering

Principal components were computed using ARPACK on the HVGs, and a kNN graph ($k = 15$) was constructed from the first 40 PCs. I ran Leiden and Louvain clustering, identifying `leiden_0.4` as the best-performing resolution (silhouette = 0.25).

ROC scoring and marker identification

An allele-tolerant marker list (*tp63.L/S*, *lef1.L/S*, *krt.L*) was used to compute a composite ROC score (`sc.tl.score_genes`). Cluster 15 showed the highest mean ROC score. Differential expression (Wilcoxon) and logistic regression (`method="logreg"`) were used to identify ROC-defining genes, yielding strong overlap (Jaccard = 0.47).

Validation and extensions

I assessed denoising (MAGIC, PCA low-rank) and batch integration (Harmony, BBKNN). Both preserved biological signal and ROC marker identity while reducing noise.

Code Availability

The full analysis pipeline, including reproducible notebooks and environment files, is available at:
github.com/lolamaturana/frog-tail-roc
and an executable version at:
https://colab.research.google.com/drive/1ZEFRZO5GRudCBl7NQhhmSN5nyY40UvkT?usp=sharing.

Results

Clustering and data structure

After quality control and normalization, the PCA–UMAP projection showed a clear organization of cell populations along developmental and regenerative axes (Fig. 1A). The Leiden and Louvain algorithms produced consistent cluster boundaries, with an Adjusted Rand Index (ARI) of 0.76 and Normalized Mutual Information (NMI) of 0.89. Most clusters corresponded to expected biological lineages such as epidermal, neural, immune, and mesenchymal populations. This confirmed that the preprocessing and dimensionality reduction steps successfully preserved biological structure while minimizing noise.

ROC identification

To locate the Regenerative Organizing Cell, I calculated a composite ROC score based on the expression of *tp63*, *lef1*, and keratin genes. The resulting UMAP heatmap revealed a single, clearly enriched epidermal cluster (cluster 15). Both Wilcoxon and logistic regression differential expression tests supported this result, showing strong overlap between their top marker sets (32 out of 50 genes, Jaccard = 0.47). The leading markers—*tp63.L/S*, *lef1.L/S*, *krt8*, and *col17a1.L*—were consistent with the regenerative epidermal phenotype described in the main paper.

Comparison with reference genes

To validate the ROC signature, I compared my top 100 ROC genes with those listed in Supplementary Table 3 from the reference study. Two genes overlapped exactly in the strict comparison ($p = 0.0027$), and five matched after collapsing `.L`/`.S` allele names (Jaccard = 0.062). Although this numerical overlap seems modest, many of the remaining genes belonged to the same keratin and transcriptional families, indicating that the recovered ROC program was biologically coherent even with minor naming discrepancies.

Denoising and integration effects

To test robustness, I repeated the analysis after data denoising (MAGIC, PCA low-rank) and after batch correction (Harmony, BBKNN). Both denoising methods slightly improved local cohesion (MAGIC silhouette = 0.31) but preserved the overall cluster topology. Harmony provided the best batch integration performance, reducing batch silhouette to –0.07 while maintaining the ROC cluster's distinct identity. Importantly, the top ROC marker genes remained stable across all versions, with over 25 of 30 overlapping between methods. This indicates that the ROC signal is not an artifact of preprocessing choices.

Skin-only embedding

To focus on the epidermal context, I re-embedded only the cells with positive skin marker scores. The resulting "skin-only" UMAP showed a compact arrangement of epidermal subtypes and a small region of high ROC scores forming a hotspot (Fig. 3). This region corresponded to the epithelial cluster that showed high *tp63* and *lef1* co-expression, consistent with the ROC population described in the original publication. Although the spatial layout of the UMAP differed slightly, the biological hierarchy of epidermal states was preserved.

Summary of findings

Altogether, the analysis reconstructed the major tissue organization of the regenerating tail, pinpointed the ROC cluster within the epidermis, and confirmed that its defining genes remain stable after multiple preprocessing and integration strategies. The high agreement between methods and the consistency with known ROC markers provide confidence that the identified population represents the same regenerative organizing cell described in prior work.

Conclusion

In this project, I reproduced the single-cell atlas of the regenerating *Xenopus laevis* tail and independently identified the Regenerative Organizing Cell (ROC) population using publicly available data. The combination of PCA–UMAP visualization, clustering, and allele-tolerant ROC scoring proved effective for recovering the key regenerative structures described in the reference dataset.

The analysis showed that the ROC is a small but distinct epidermal population characterized by co-expression of *tp63*, *lef1*, and keratin genes. Its marker set overlapped substantially with that of the original study, and its localization remained stable across denoising and batch-correction methods, supporting the biological validity of this population.

Overall, these results demonstrate that an open, reproducible workflow based on Scanpy can accurately recover fine-grained regenerative cell types in complex multi-batch single-cell data. The full notebook and report provide a transparent framework that can be extended to other regenerative models or future cross-species analyses of cell fate and tissue repair.

References

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